

COM S 127 - Lab Week 13 Grading Rubric

This lab was assigned for week 13 of the Fall semester.

This lab is due by 11:59 p.m. of the sixth day after the student's initial lab meeting day. Meaning, for example, if a student has their lab on **THURSDAY**, they would have until the following **WEDNESDAY** to complete their work.

Lab Objective

The purpose of this lab is for students to begin working with converting decimal (base ten) numbers to binary (base two) and binary (base two) numbers to decimal (base ten).

Instructions

Submission 'Check Offs'

For students to receive credit for their lab work, they must 'check off' their exercises with either the Graduate TA or Undergraduate TA (UGTA) for their lab section. If the lab cannot be completed during the assigned lab hours, the student **MUST** either attend a **Graduate TAs/ Undergraduate TAs** office hours to complete the 'check off' process, or arrange for a Zoom/ WebEx meeting with either a Graduate TA or an Undergraduate TA to receive credit for their work.

This means that **STUDENTS WILL NO LONGER BE ABLE TO SUBMIT THEIR WORK ON CANVAS**. As such, **all work MUST 'checked off' in person or via a Zoom/ WebEx appointment BEFORE the student's next lab meeting**. This means that students will no longer be able to send an email about completed work *before* the deadline, and then check off that work *after* the deadline.

For clarity: **ALL WORK MUST BE CHECKED OFF BEFORE THE FINAL DEADLINE**.

There is a list of Graduate TA office hours and email addresses at the top of the syllabus on Canvas.

There is a list of Undergraduate TA office hours and email addresses at the top of the syllabus on Canvas.

Important Notes:

All scripts should include the student's name and the date they programmed the script on the top line of the file. The student should also include the week of the lab, and the exercise number on the second line. Example:

```
# <Student Name>                <The Date>
# Lab Week <week of the lab> - Exercise #<exercise number>

# Matthew Holman                11-14-2022
# Lab Week 13 - Exercise #1
```

Readings:

https://en.wikipedia.org/wiki/Binary_number

- Counting in binary
- Binary arithmetic
 - Addition
 - Subtraction
- Conversion to and from other numeral systems
 - Decimal to Binary (**HINT**: Watch the animation **very** closely - what number represents the most significant bit in the pattern? What makes it different from all the other numbers in the pattern the way it is constructed? Is this number a remainder like all the others? If not, what would you have to do to have it be a remainder like all the others?)
 - Binary to Decimal

Lab Challenge Activities:

- **Create a script** called **exercise1.py**, such that it completes the task in **Exercise 1** of the **labWeek13Binary.pptx** file.
- **Create a script** called **exercise2.py**, such that it completes the task in **Exercise 2** of the **labWeek13Binary.pptx** file.

Deliverables:

- **Verbally explain all the concepts in the Readings section of this document to a Graduate TA/ UGTA without looking at the website, such that the Graduate TA/ UGTA is confident you fully understand these concepts.**
- **exercise1.py** which completes the tasks laid out in the **Lab Challenge Activities** section of this document.
- **exercise2.py** which completes the tasks laid out in the **Lab Challenge Activities** section of this document.
- **All the deliverable scripts should be shown to a Graduate TA/ UGTA**

Resources:

- https://en.wikipedia.org/wiki/Binary_number

Files Provided

- labWeek13Binary.pptx

Example Code/ Output

Exercise 1:

Enter an Integer: 5
The binary representation of 5 is 101

Enter an Integer: 17
The binary representation of 17 is 10001

Exercise 2:

Enter a (Binary) String: 101
The decimal representation of 101 is 5

Enter a (Binary) String: 10001
The decimal representation of 10001 is 17

Grading Items

- **(Attendance)** Was the student present at the lab or had made arrangements to attend virtually?: _____ / 5
- **(Reading)** Has the student demonstrated that they have completed their reading by verbally explaining the concepts in the **Reading** section to a Graduate TA/ UGTA?: _____ / 15
- **(exercise1.py)** Did the student successfully complete a script which generates correct output as per the **Lab Challenge Activities** section?: _____ / 5
- **(exercise2.py)** Did the student successfully complete a script which generates correct output as per the **Lab Challenge Activities** section?: _____ / 5

TOTAL _____ / 30